Machine Learning 1

Super AI Engineer Season 6

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Presented by

Aslan B. Wong

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REGRESSION ANALYSIS IN DATA SCIENCE

PROJECT CARD & KEY LEARNING OUTCOMES

PROJECT CARD

GOAL:

- *Build, train, and test an XG -Boost model to perform regression tasks and predict chances of university admission given student's profile.*

TOOL:

- *Jupyter Notebooks, Google Colab and Scikit -Learn*

PRACTICAL REAL -WORLD APPLICATION:

- *Determine top qualifying students by University Admission Office.*

DATA:

INPUTS (FEATURES):

- *GRE Scores (out of 340)*
- *TOEFL Scores (out of 120)*
- *University Rating (out of 5)*
- *Statement of Purpose (SOP)*
- *Letter of Recommendation (LOR) Strength (out of 5)*
- *Undergraduate GPA (out of 10)*
- *Research Experience (either 0 or 1)*

OUTPUTS:

- *Chance of admission (ranging from 0 to 1)*

- Data Source: https://www.kaggle.com/mohansacharya/graduate-admissions
- Photo Credit: https://www.pexels.com/photo/accomplishment-ceremony-education-graduation-267885/

MODEL INPUTS & OUTPUTS

INPUTS

GRE Score
TOEFL Score
University Rating
Statement of Purpose (SOP)
Letter of Recommendation (LOR)
CGPA
Research

XG-BOOST REGRESSION MODEL

OUTPUT

Chance of Admission

KEY LEARNING OUTCOMES

Perform data visualization using Seaborn & Matplotlib libraries

Train an XG -boost algorithm to predict university admission

Train an XG -boost algorithm to predict life expectancy (capstone project)

Understand the theory/intuition behind boosting

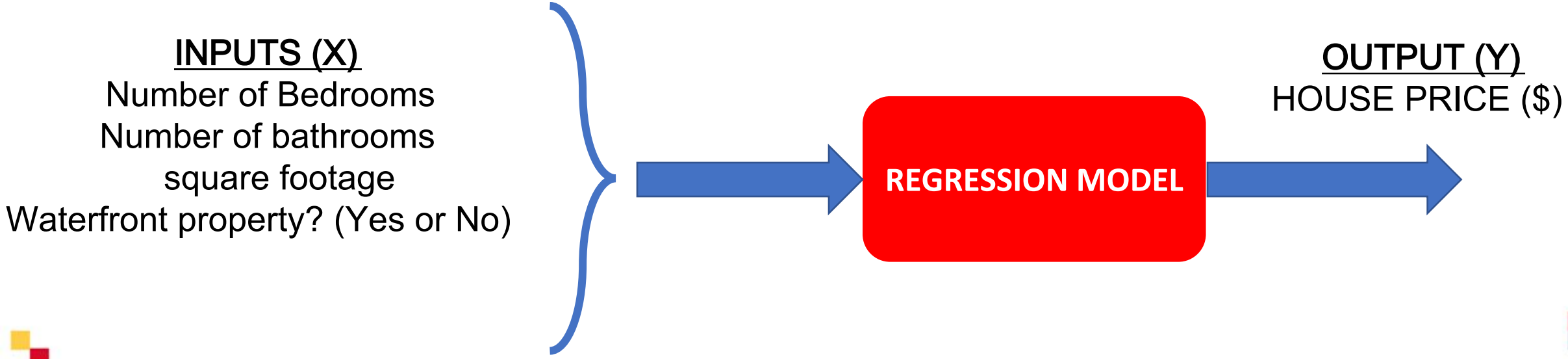
Learn the advantages & disadvantages of XG-boost

Evaluate trained models performance

WHAT IS REGRESSION?

WHAT IS REGRESSION?

- Regression works by estimating a continuous dependent variable Y from a list of independent input variables X.
- Regression is used in many real-life applications:
  - Financial forecasting
  - Weather analysis



WHAT IS XGBOOST?

& WHY?

WHAT IS XGBOOST? & WHY?

- XGBoost or Extreme gradient boosting is the go-to algorithm for many data scientists.
- XGBoost is a supervised learning algorithm that implements gradient boosted trees algorithm.
- The algorithm works by building a series of models from the training data, each model is built to correct the mistakes made by the previous model. Models are added sequentially until no further improvement can be made.

"When in doubt, use xgboost!" – Owen Zhang, 1st place Kaggle Winner

Extremely fast

Good memory utilization

Works for regression & Classification

Robust

PROJECT DEMO

PROJECT DEMO

Regression Analysis in Data Science.ipynb

File Edit View Insert Runtime Tools Help

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Files

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RAM Disk Editing

- sample_data
- Life_Expectancy_Data.csv
- university_admission.csv

CODING TASK #1: IMPORT LIBRARIES/DATASETS

[3]
```
import seaborn as sns
import pandas as pd
import matplotlib.pyplot as plt
```

[4]
```
# Read the CSV file
university_df = pd.read_csv("university_admission.csv")
```

[5]
```
# Load the top 6 instances
university_df.head(6)
```

	GRE_Score	TOEFL_Score	University_Rating	SOP	LOR	CGPA	Research	Chance_of_Admission
0	337	118	4	4.5	4.5	9.65	1	0.92
1	324	107	4	4.0	4.5	8.87	1	0.76
2	316	104	3	3.0	3.5	8.00	1	0.72
3	322	110	3	3.5	2.5	8.67	1	0.80
4	314	103	2	2.0	3.0	8.21	0	0.65
5	330	115	5	4.5	3.0	9.34	1	0.90

FINAL CAPSTONE PROJECT

FINAL CAPSTONE PROJECT

- In this hands-on project, we will train an XG-Boost model to predict life expectancy.
- The data is obtained from World Health Organization (WHO) and United Nations Website and contains features like year, status, life expectancy, adult mortality, infant deaths, percentage of expenditure, alcohol etc.

INPUT FEATURES

YEAR
ADULT MORTALITY
STATUS
INFANT DEATHS
ALCOHOL
HEPATITIS B

→ **XG-BOOST REGRESSION MODEL** →

PREDICTION

LIFE EXPECTANCY

Source: https://www.kaggle.com/jkumarajarshi/life-expectancy-who

FINAL CAPSTONE PROJECT TASKS

Final Capstone Project Tasks:

1. Import the "*Life_Expectancy_Data.csv*" dataset using Pandas
2. Check if missing values exist in the data, perform feature engineering to eliminate or fill missing values
3. What is the memory usage of the DataFrame
4. List the minimum, average and maximum life expectancy
5. Plot the histogram, pairplot and correlation heatmap for all features
6. Plot the scatterplot between "Income composition of resources" and "life expectancy", use "status" for hue attribute. Comment on the plot
7. Plot the scatterplot between "Schooling" and "life expectancy", use status for hue attribute. Comment on the scatterplot.
8. Split the data into 80% for training and 20% for testing
9. Train an XG-boost algorithm
10. Assess trained regression model performance, what is R2?
11. Plot trained model predictions vs. ground truth output